Household Animals Classification Using Deep Learning

Team details-

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1. Introduction

This project classifies an input image as Cat or Dog. Three approaches were compared: a VGG-like convolutional neural network (CNN) trained from scratch; a regularized CNN using random horizontal flips and dropout; and ImageNet-pretrained VGG16 using transfer learning followed by fine-tuning. Images were resized to 224 × 224 pixels and models were trained with binary cross-entropy and RMSprop.

Dataset and preprocessing

The implementation used the Microsoft Cats vs Dogs archive because the Kaggle competition download was unavailable in the working environment. After validation, 12,499 cat and 12,499 dog images were retained (24,998 total). Each class was split into 9,999 training, 2,000 validation, and 500 test images, approximately an 80/16/4 split. Images were converted to RGB JPEGs to avoid decoder errors and loaded in batches of 32. This source differs from the Kaggle dataset described in the reference project report.

2. Architecture and training

VGG-like CNN: 3×3 convolutional layers with 32, 64, 128 and 128 filters, ReLU activations and max pooling, followed by Flatten, Dense(128) and sigmoid output. **Regularized CNN:** the same backbone with horizontal-flip augmentation and 0.5 dropout. **VGG16 transfer:** pretrained convolutional layers were frozen while a Global Average Pooling and Dense(128) sigmoid head was trained. Fine-tuning then unfroze layers from block5_conv1 using a low learning rate.

3. Results

Model	Test loss	Test accuracy
VGG-like baseline	1.6643	85.10%
Dropout + augmentation CNN	0.1856	92.20%
VGG16 transfer (frozen)	0.1036	98.40%
VGG16 fine-tuned	0.0951	98.60%

The fine-tuned VGG16 achieved the best reported test accuracy (98.60%), 0.20 percentage points above the frozen transfer model. The scratch baseline's training accuracy rose while validation loss increased late in training, indicating overfitting. Dropout and augmentation improved test performance relative to the baseline.

4. Visualizations and conclusion

Learning curves compare training and validation accuracy and loss. Intermediate convolutional activations show feature responses at different depths. Grad-CAM provides a coarse qualitative heatmap of regions influencing the selected class score. The provided example is predicted as Cat, with dog probability 0.0000; this visualization is illustrative and does not prove the model relies only on relevant features.

Transfer learning was the strongest approach in this experiment, reaching 98.60% test accuracy. Further work should include a confusion matrix, precision/recall/F1, checks for duplicates or dataset bias, and testing on independently collected images. Performance is specific to this split and run and is not guaranteed for all household-animal images.

Reference basis: Lei Lin, "Household Animals Classification Using Deep Learning," CS229 Spring 2020 project report. Dataset in this implementation: Microsoft Cats vs Dogs archive.